

QUIZ 3 REVIEW (CH3 & ARITHMETIC)

OLD QUIZ 3

1) Find the multiplicative inverse

a) $5^{-1} \equiv \underline{9} \pmod{11}$

b) $43^{-1} \equiv \underline{47} \pmod{101}$

$$\begin{bmatrix} 43 & 1 \\ 101 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 43 & 1 \\ 101 & 0 \end{bmatrix} = \begin{bmatrix} 101 & 0 \\ 43 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} 101 & 0 \\ 43 & 1 \end{bmatrix} = \begin{bmatrix} 43 & 1 \\ 15 & -2 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} 43 & 1 \\ 15 & -2 \end{bmatrix} = \begin{bmatrix} 15 & -2 \\ 13 & 5 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 15 & -2 \\ 13 & 5 \end{bmatrix} = \begin{bmatrix} 13 & 5 \\ 2 & -7 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & -6 \end{bmatrix} \begin{bmatrix} 13 & 5 \\ 2 & -7 \end{bmatrix} = \begin{bmatrix} 2 & -7 \\ 1 & 47 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} 2 & -7 \\ 1 & 47 \end{bmatrix} = \begin{bmatrix} 1 & 47 \\ 0 & -91 \end{bmatrix} \leadsto \textcircled{47}$$

2) Suppose $d = \gcd(a, m)$ and $ab \equiv ac \pmod{m}$. Show that $b \equiv c \pmod{\frac{m}{d}}$

$$ab \equiv ac \pmod{m} \Rightarrow m \mid ab - ac = a(b - c) \Rightarrow \frac{m}{d} \mid \frac{a}{d}(b - c)$$

$$d = \gcd(a, m) \Rightarrow \exists x, y \in \mathbb{Z}, \text{ s.t. } d = ax + my \Rightarrow 1 = \frac{a}{d}x + \frac{m}{d}y \Rightarrow b - c = \frac{a}{d}(b - c)x + \frac{m}{d}(b - c)y$$

$$\therefore \frac{m}{d} \mid \frac{a}{d}(b - c) \Rightarrow \frac{m}{d} \mid \frac{a}{d}(b - c)x \Rightarrow \frac{m}{d} \mid \frac{a}{d}(b - c)x + \frac{m}{d}(b - c)y = b - c \Rightarrow b \equiv c \pmod{\frac{m}{d}} \quad \square$$

3) The first 9 digits $x_1, x_2, \dots, x_9$, of an ISBN number are already chosen. Then x_{10} is defined by $x_{10} = \underline{\sum_{i=1}^9 i x_i \pmod{11}}$ Don't forget